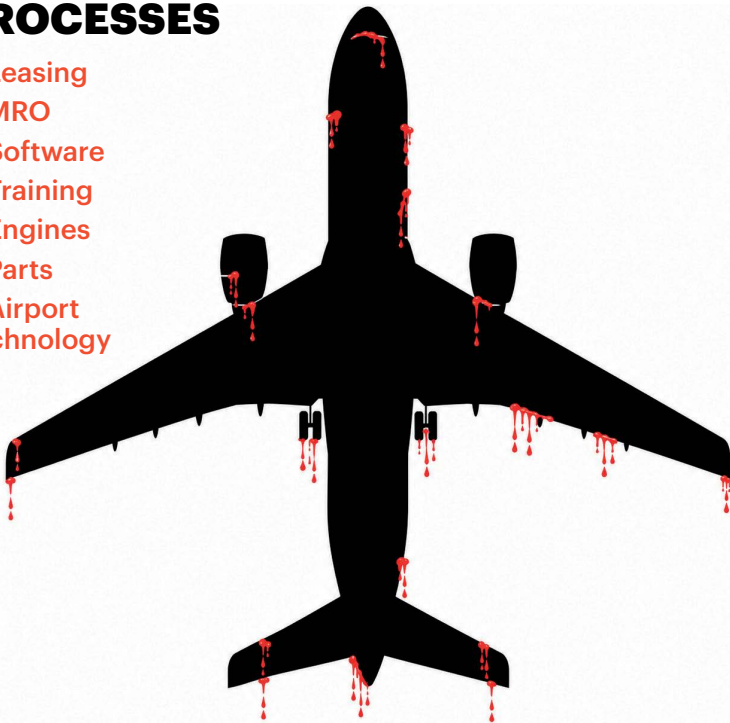


IMPORT-DEPENDENT AVIATION PROCESSES

- Leasing
- MRO
- Software
- Training
- Engines
- Parts
- Airport technology



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dependency in depth, identifies the material constraints that underpin this reality. "India currently lacks adequate expertise in manufacturing aerospace-grade materials — especially titanium and aluminium superalloys, as well as advanced composite materials used in aircraft structures," he says.

He explains that these materials require high precision, quality control, and specialised industrial ecosystems, which are still evolving domestically.

Karthik Kudkuli, Assistant Professor of Aviation and Aerospace Management at CMS Business School, JAIN (Deemed-to-be University), frames the systemic gap. According to him, India is not short of aerospace talent, but it still lacks a fully mature end-to-

end aviation ecosystem. The real issue is fragmented capabilities in R&D, test infrastructure, certified suppliers, materials, and industrial coordination.

The HAL Hindustan-228 — a 19-seater turboprop certified under a licence from Germany in the 1970s — is India's only commercially certified, domestically manufactured aircraft. The airline that operated its first scheduled commercial flight, Alliance Air, remains a state-owned regional carrier. Even the Hindustan-228's engine is foreign: a Honeywell Garrett TPE331, designed in the United States.

A committed order book extends this import dependence for a generation. Air India has orders for 570 aircraft from Airbus and

Boeing. IndiGo has 920 on order, the largest single-airline backlog in the world. Akasa Air has placed an order for 226 Boeing 737 MAXs. The three carriers' backlogs together total roughly 1,716 aircraft with a combined discounted value of \$88–107 billion, every rupee of which flows to Toulouse and Seattle.

Melligeri is candid about the ceiling. "A fully indigenous aircraft is unlikely," he says. "At best, it will be an 'assembled in India' aircraft with design, engines, and key systems sourced from global OEMs and partners, perhaps through joint ventures." India currently accounts for about 2% of the global aerospace components manufacturing market, with the industry targeting a 10% share. The ambition is real. The distance is very large.

LEASING ARCHITECTURE

Of India's 860-plus commercial aircraft, 86–87% are leased from foreign entities—compared with a global average of around 53%. The leasing industry is concentrated in Dublin, where AerCap, the world's largest lessor with a portfolio of 1,669 aircraft valued at nearly \$62 billion, maintains its headquarters alongside Avolon and SMBC Aviation Capital. Together, these three companies controlled approximately 30% of the global commercial fleet in 2024, according to Mordor Intelligence.

The IMPRI Impact and Policy Research Institute estimates that India's dependence on offshore leasing costs it "in excess of \$5 billion annually in foreign exchange." Group Captain Chari puts the overall industry annual forex outflow—leasing, maintenance, repair, and operations (MRO), software, parts, and pilot